BUILDINGS AS GRAPHS

MaCAD Digital tools for Graph Machine Learning seminar

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WASSIM JABI

Wassim Jabi is Course Director of the MSc Computational Methods in Architecture at the Welsh School of Architecture, Cardiff University. He earned his M.Arch. and PhD from the University of Michigan and taught in the United States before moving to the UK in 2008, where he secured a National Science Foundation grant as Principal Investigator. His 30+ years of research experience spans parametric and generative design, spatial representation, building performance simulation, machine learning, robotic fabrication, and graph-based methods. He has particular expertise in graphs, graph analysis, and graph machine learning applied to the built environment. He authored Parametric Design for Architecture (Laurence King). A Leverhulme Trust grant led to the development of the Topologic software library.

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OLGA POLETKINA

Olga Poletkina is an architect and BIM specialist with over 15 years of experience. With a background in traditional architectural design, she gradually shifted her focus to digital technologies and automation in the building industry. Her interest in data science and artificial intelligence led her to explore how machine learning can be applied to architectural workflows. Today, she combines practical expertise with research, aiming to integrate advanced computational methods into the design and construction process.

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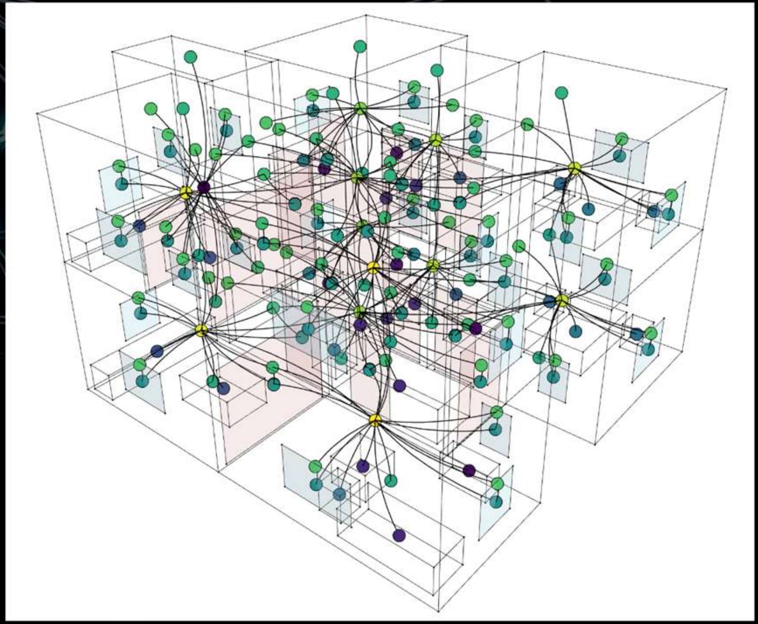
WHAT IS THIS SEMINAR ABOUT?

This seminar explores the proposition that architecture is fundamentally relational. Buildings and cities are not merely collections of objects; they are structured fields of spatial, functional, environmental, and social relationships. Graphs provide a rigorous and computationally tractable way of making these relationships explicit. The course equips participants with the conceptual and technical tools required to represent architectural and urban systems as graphs and to apply graph machine learning (GML) methods to extract insight from them.



LEARNING OBJECTIVES

- Learn the fundamentals of Graph Theory
- Learn the fundamentals of Graph Machine Learning
- Learn how to represent buildings as graphs
- Learn how to work with graph databases
- Explore different applications of GraphML

COURSE STRUCTURE

1. Architecture as Relational Structure: An Introduction to Graph Thinking in Design (09.04.26)
2. From Geometry to Topology: Representing Buildings as Graphs (09.04.26)
3. Spatial Intelligence: Graph Analysis Methods for Architectural Analysis (14.04.26)
4. From 3D Models to Attributed Graphs: Extracting Structure from IFC and Computational Geometry (21.04.26)
5. Designing Graph Datasets: Feature Engineering, Encoding, and Data Strategy for the Built Environment (29.04.26)
6. Graph Machine Learning for Architects: Message Passing, Hyperparameters, and the End-to-End Workflow (19.05.26)
7. Graph Generation and Graph DB Neo4j (02.06.26)

SOFTWARE ENVIRONMENT

1. Google Colab OR Microsoft Visual Studio Code with Python 3.11.
2. TopologicPy (pip install topologicpy)
3. Rhinoceros 5.0 or 6.0. The 90-day trial version can be downloaded from the website www.rhino3d.com/eval.html
4. Adobe Suite or equivalent open-source/free
5. *Anything else you need!*

ASSIGNMENTS

1. *Consult the Syllabus Document*

DELIVERABLES

1. Creation of the seminar's blog post:
2. Students are requested to submit all the material on the IAAC Gdrive and a Blog post needs to be curated on iaacblog.net for each project (see "IAAC | Publication Guidelines" https://drive.google.com/open?id=0B6cIj8JAKs2rVTNYNEg2c093M2s) within a maximum of 2 days after the end of the Seminar.
3. Students who submit after the deadlines defined by the faculty and coordination will be subject to
4. penalty and the grade will be automatically lowered by 0.3 point for every day of delay.
5. Incomplete submission is considered as missing submission.

GRADING SYSTEM

- 0 - 4.9 Fail (submission of a supplementary work by June)
- 5.0 -6.9 Pass
- 7.0 - 8.9 Good
- 9.0 - 10 Excellent/Distinction.

On this basis, students will be evaluated on

several aspects such as:

1. Attendance: 15 %
2. Results: 45 %
3. Presentations: 20 %
4. Blog posts: 20 %

REFERENCES / BIBLIOGRAPHY

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- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C. & Yu, P.S., 2021. A comprehensive survey on graph neural networks. IEEE Transactions on Neural Networks and Learning Systems, 32(1), pp.4–24. doi:10.1109/TNNLS.2020.2978386.
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- Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, E., Mody, A., Truitt, S. & Larson, J., 2024. From local to global: A graph RAG approach to query-focused summarization. arXiv preprint. doi:10.48550/arXiv.2404.16130.
- *And Much more!....*

QUESTIONS?

LET'S GET STARTED!